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High-accuracy estimation method of typhoon storm surge disaster loss under small sample conditions by information diffusion model coupled with machine learning models

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ABSTRACT

Typhoon storm surge disaster is the most severe marine disaster in China. Accurate estimation of typhoon storm surge disaster loss (TSSDL) is significant for emergency decisions and economic sustainability development. However, the TSSDL estimation is limited by small sample conditions, resulting in the low accuracy of the TSSDL estimation model based on machine learning. To solve the problems of easy overfitting and poor generalization ability of machine learning models under small sample conditions, the high-accuracy TSSDL estimation method was proposed. Firstly, this estimation method combines Gaussian Noise with the Information Diffusion Model based on the Vibrating String equation (named GN-VSIDM) to generate virtual samples to augment the original training set. Then, the augmented training set was applied to train three machine learning models. The results are as follows: the virtual sample generation method, i.e., GN-VSIDM, solves the small sample problem in the TSSDL estimation process and improves the machine models' accuracy and robustness. Based on the GN-VSIDM and the eXtreme Gradient Boosting (named XGBoost) methods, the joint model GN-VSIDM-XGBoost is the optimal TSSDL estimation model. Compared with the original XGBoost model, the RMSE and R^2 of the GN-VSIDM-XGBoost model are 0.1089 and 0.8292, which reduces 25.67% and improves 19.88%, respectively. Besides, the GN-VSIDM-XGBoost model possesses excellent robustness. The GN-VSIDM overcomes the limitation on the performance of TSSDL estimation models based on machine learning under small sample conditions. This study provides an effective case and method for solving the small sample problem in disaster loss assessment.

1. Introduction

A typhoon storm surge is an abnormal rise of sea level caused by tropical cyclones [1]. Due to the low pressure at the tropical cyclones' center, seawater is drawn upwards, making the sea level higher than normal. At the same time, the powerful driving effect caused by tropical cyclones also pushes plenty of seawater toward the coast [2]. Once the increased coastal seawater exceeds the warning tidal level, the coastal defense facilities and buildings would be damaged, vast areas of farmland would be flooded, mass people would be hurt, and the economy would suffer a severe loss [3,4]. Among numerous marine disasters affecting China, the typhoon

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storm surge disaster (TSSD) is the most destructive and causes the most severe socio-economic losses [5,6]. According to the official statistics, the typhoon storm surge disaster loss (TSSDL) reached RMB 218.5 billion from 2000 to 2020 in China, accounting for 92% of the total losses of marine disasters [7]. As China's marine economy continued to grow between 2016 and 2019, the proportion of Gross Ocean Product in the coastal area's Gross Domestic Product (GDP) increased from 16.4% to 17.1% [8]. However, the development of the marine economy leads to a large influx of population, and industrial agglomeration, which significantly increases the exposure of disaster-bearing bodies and infrastructure vulnerability in coastal areas [9,10]. Thus, the risk of TSSD rises along with economic development, which might cause more severe TSSDL. Moreover, as proposed in the Sendai Framework for Disaster Risk Reduction, reducing direct economic losses from disasters has become the focus and goal of global disaster prevention and mitigation [11]. Besides, the high accuracy TSSDL estimation model can provide relevant departments with reliable and timely disaster prevention and mitigation decision support [12]. However, the small sample conditions, i.e., the insufficient number and incomplete information of post-disaster loss sample statistics, limit the accuracy of the loss estimation model [13]. Therefore, under the condition of small TSSDL samples, constructing a TSSDL estimation model with high accuracy and strong robustness is of great academic value and practical significance.

Many loss estimation methods for marine and storm surge disasters have been researched by scholars. Overall, the estimation models are mainly divided into physical models and machine learning models (MLMs) [14]. The physical models estimate the economic losses of TSSDs based on the quantification of the inundated area, submerged depth, inundated elements, and the degree of exposed parts being direct tangible damage [15]. For example, the Sea, Lake and Overland Surges from Hurricanes (SLOSH) model applies typhoon parameters data and coastal sounding elements data to estimate storm surge heights [16,17]; the Advanced Circulation and Simulating Waves Nearshore (ADCIRC-SWAN) coupled model uses the wind field data to simulate the storm surge scenario [15]. By SLOSH or ADCIRC-SWAN, the inundated area and submerged depth can be quantified. Thus, the economic loss of the storm surge disaster can be estimated by combining the inundated area and submerged depth with the loss functions. Besides, the flood model in HAZards U.S. MultiHazard (HAZUS-MH) contains built-in loss functions for various types of buildings, which estimates the loss of buildings caused by storm surge disasters under different inundation scenarios [18]. The physical models based on the physical mechanisms and statistical data can calculate a relatively reliable result. However, because many historical statistics and expensive computing resources are challenging to obtain, the physical models achieve poor real-time estimation performance [19]. Besides, the physical model is limited to a specific study area with poor applicability.

The MLM is characterized by parallel computing, universal adaptability, and strong robustness, which provides better real-time performance and saves more computing resources for TSSDL estimation. Furthermore, the MLM shows an excellent performance in TSSDL estimation with robust analysis and processing capabilities for high-dimensional and nonlinear data [14]. For example, a variety of MLMs, such as BPNN [20–22], RF [13,23,24], and Support Vector Machine (SVM) [4,10,25], are widely used in TSSDL estimation and achieves excellent results. Therefore, it has advantages and mature research bases to estimate the TSSDL by MLMs. Different from algorithms based on a single learner, ensemble learning is a learning strategy that combines multiple learners to obtain superior generalization ability and achieves excellent performance in research tasks based on small sample datasets [13,26,27]. Thus, ensemble learning is potential for estimating the TSSDL and three widely-used MLMs based on ensemble learning, i.e., LSBoost, XGBoost, and RF, are applied to estimate the TSSDL [12,13,28,29].

Due to the small number of TSSDs occurring each year and difficulties in data collection such as high costs, time constraints, and lack of technology, the TSSDL data is insufficient, and the information is incomplete. However, a large number of disaster loss statistics are needed to train and optimize the TSSDL estimation model to achieve excellent performance with high accuracy and strong robustness. Therefore, the small sample size and information incompleteness lead to large errors and poor performance of the TSSDL estimation models, which is called the small sample problem [30]. Furthermore, driven by data, the model parameters of an MLM are necessarily adjusted by learning a large amount of historical data to minimize the error. Inevitably, the small sample problem creates limitations on the TSSDL estimation models based on machine learning, such as easy overfitting and poor generalization ability. Consequently, this paper focuses on solving the constraints caused by the small sample problem during the construction of the TSSDL estimation models based on the MLMs.

To handle the small sample problem in machine learning, the three methods, i.e., gray prediction model, feature extraction, and virtual sample generation (VSG), are mainly used in current studies. Among the above methods, VSG, proposed by Niyogi et al. [31], is the most promising and popular method [32]. The strategy of VSG is to extract prior information from the given small sample data and then generate new virtual samples to fill the information gaps among the original samples [33]. Many VSG methods are applied in marine natural disaster loss estimation studies to effectively solve the small sample problem. For example, Sun et al. overcame the data scarcity problem through the VSG method, namely the k-nearest neighbor-Gaussian noise method, which adds Gaussian noise to original samples to generate virtual samples [13]. Zhao et al. verified that adding virtual loss data by interpolation to generate virtual samples can effectively solve the small sample problem [34]. Li et al. solved the small sample problem through the information diffusion model (IDM), which calculates virtual disaster loss data to generate virtual samples [35]. Consequently, the VSG methods show excellent performance and effectiveness in solving the small sample problem.

Among methods of VSG, the IDM has advantages such as systematic theoretical derivation and a large number of practical applications. The IDM, proposed by Huang [36], is a fuzzy statistical technique that can transform a single-point sample into a set-value sample. It effectively utilizes fuzzy information in small samples to fill in the information gaps and compensate for the information deficiency [36]. The IDMs are widely used in natural disaster risk analysis and achieve excellent performance, such as earthquakes [37], floods [38], soil erosion [39], and agriculture drought [40]. The IDM extracts more prior information from incomplete small samples and reasonably infers missing data by approximate inference methods. Thus, the virtual samples generated by the IDM possess similar data distribution and statistical characteristics as the original samples. Therefore, the IDM has excellent research value and practical potential in TSSDL estimation under small sample conditions.

Moreover, based on the simplest normal IDM [38], a variety of improved models are proposed, such as the Cloud-based Information Diffusion (CID) model [41], Diffusion Function based on Vibrating String equation (VSDF) [42], Information Diffusion Model based on Genetic algorithm (GIDM) [43], Information Diffusion Model based on Variable Fuzzy Set (VSF-IDM) [38]. These improved models achieve better performance in analyzing different disaster risks. The IDM based on the vibrating string equation (named VSIDM) effectively extracts the prior information of the natural disaster data that follows relatively complex distribution patterns. This model achieves excellent performance on earthquake observation data in Yunnan Province [42] and marine disaster data in Shanghai [35]. Consequently, the VSIDM is applied in this paper to solve the small sample problem by generating virtual samples during TSSDL estimation.

In summary, the high-accuracy estimation method of TSSDL, which combines the IDM and MLMs, was proposed to achieve the goal of accurate estimation of TSSDL under small sample conditions. This estimation method includes the following five parts. Firstly, an objective and comprehensive TSSDL estimation indicator system was constructed. Secondly, the random forest-recursive feature elimination method based on k-fold cross-validation (RF-RFECV) was used to screen the indicators to reduce the model complexity and indicator redundancy. Thirdly, a new augmented training set was generated through the proposed VSG method, which combines Gaussian Noise with the Information Diffusion Model based on the Vibrating String equation (named GN-VSIDM). Then, cross-contrast experiments with various MLMs and different VSG methods were conducted to obtain the optimal joint model of TSSDL estimation. Finally, the robustness of the optimal model was verified.

The remainder of this paper is organized as follows. Section 2 introduces the details of research materials and experimental methods. Section 3 presents the experimental results and their analysis. Section 4 discusses the research results. Section 5 offers the conclusion.

2. Materials and methods

The materials and methods of this paper mainly include the following parts: data processing, constructing a comprehensive TSSDL estimation indicator system, screening the indicators by the RF-RFECV method, generating a new augmented training set by the GN-VSIDM to build the three MLMs to estimate TSSDL, evaluating the optimal model's performance and verifying its robustness. As shown in Fig. 1, the materials processing and methods implementation of TSSDL estimation is divided into nine steps to accomplish in detail.

2.1. Study area

The southern marine economic circle is chosen as the study area, consisting of four provincial-level administrative regions. As shown in Fig. 2, the four provincial-level administrative regions are Guangdong (109°45′–117°20′E, 20°09′–25°31′N), Guangxi (104°28′–112°04′E, 20°54′–26°23′N), Hainan (Hainan Island, 108°37′–111°03′E, 18°10′–20°10′N), and Fujian (115°50′–120°40′E, 23°33′–28°20′N). The study area is one of the most significant economic development regions in China, with a dense population and developed economy. In 2020, the total population and regional GDP in this area accounted for 16.16% and 18.08% of the nation's, respectively [44]. The marine economy develops continuously in this area, and the Gross Ocean Product value reached RMB 3092.5 billion (in 2020), accounting for 38.7% of the nation's [8]. However, under a high incidence of tropical cyclones, the region suffered from severe TSSDL from 2000 to 2020, accounting for 68.69% of the total TSSDL in China on average, as shown in Fig. 3. In addition, as can be seen from Fig. 3, there is a large deviation in the percentage of losses in the study area in different years, which is closely related to the landfall location of the tropical cyclones that caused the severe losses [7]. For example, in 2018, the tropical cyclones that caused severe losses, namely Maria and Mangkhut, made landfall and caused huge losses in the study area. In 2019, the tropical cyclones that caused severe losses, namely Lekima and Mitag, made landfall in Zhejiang Province and caused little damage to the study area. In different years, there is some randomness in the landfall locations of tropical cyclones that cause severe losses, which leads to biases in the percentage of losses in the different years in the study area. Zhejiang Province (118°01′–123°10′E, 27°02′–31°11′N) is chosen as the verification area, which is close to the study area and TSSDs occur frequently. Besides, the verification area also suffered from severe TSSDL from 2000 to 2020, accounting for 17.66% of the total TSSDL in China on average.

2.2. Data sources and processing

2.2.1. Data sources

Official data were collected to construct an indicator system for ensuring the accuracy of the TSSDL estimation model. The data source, content, and purpose are described as follows.

(1) TSSDL statistics

The statistical data of TSSDL are provided by the *Collection of Storm Surge Disaster Historical Data in China 1949–2009* [45] and *Bulletin of China Marine Disaster* [7], including maximum water increase, ultra-warning water level, affected population, disaster area of mariculture, submerged farmland area, Coastal facilities (embankments and berms) and damaged fishing boats. TSSDL data reflects the danger and damage degrees of typhoon storm surge.

(2) Typhoon statistics

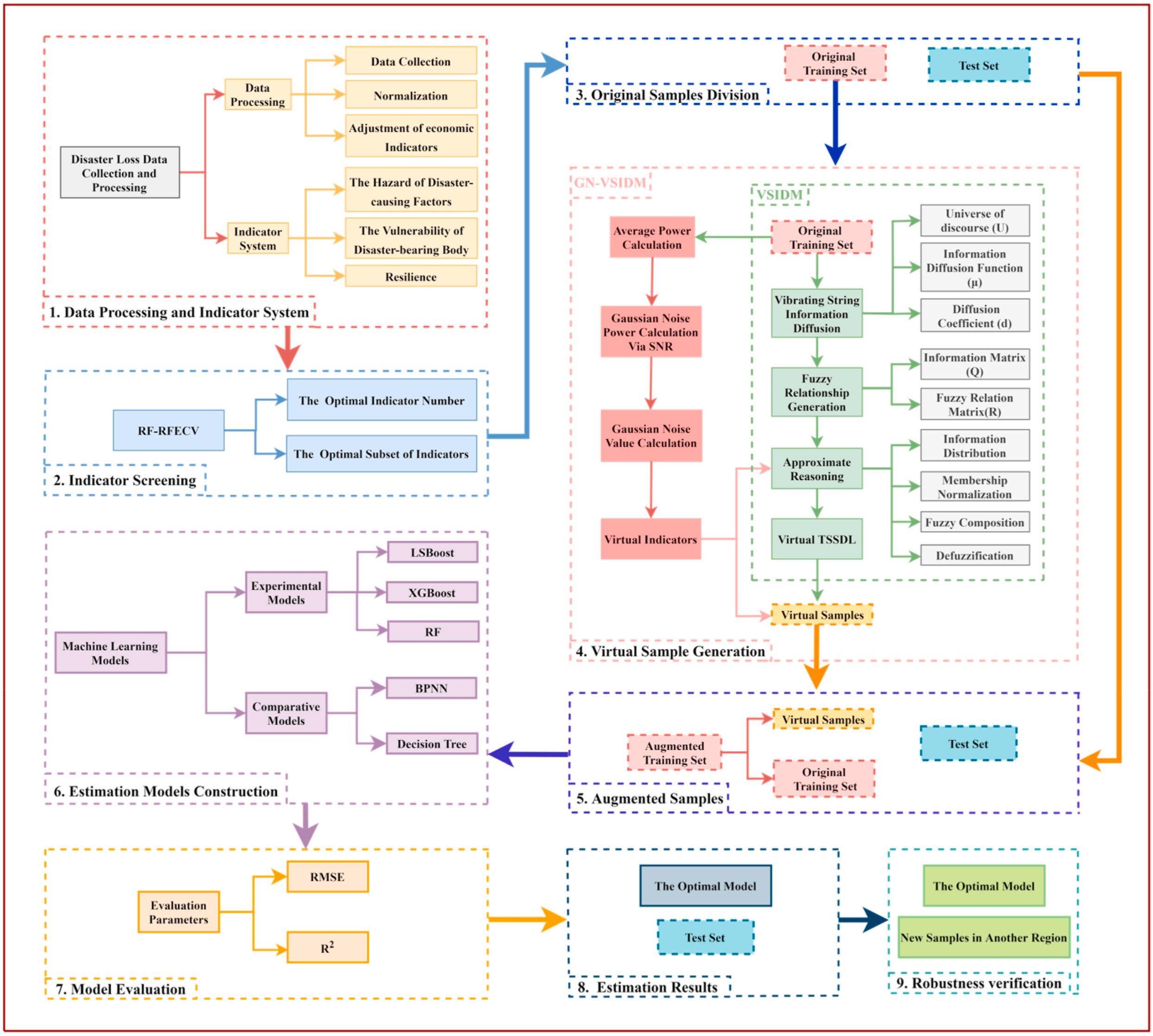

Fig. 1. The research framework diagram of TSSDL estimation.

The typhoon track information is obtained from *CMA-STI Best Track Dataset for Tropical Cyclones over the western North Pacific* (https://tcdata.typhoon.org.cn/) [46,47], *Zhejiang Province Ocean Forecasting Website* (https://zjocean.org.cn/typhoon) and *National Meteorological Centre Typhoon Website* (http://typhoon.nmc.cn/web.html).

Typhoon statistics, including maximum wind speed, central pressure, forward speed, typhoon tracks, and grades of tropical cyclones, are collected and divided into two sub-datasets. The typhoon tracks of the study area (52 typhoon statistics) and Zhejiang Province (26 typhoon statistics) are shown in Fig. 4 (a) and Fig. 4 (b), respectively. Typhoon statistics in the study area are used to construct the indicator system, while typhoon statistics in Zhejiang Province are used to verify the robustness of the TSSDL estimation model.

(3) Socioeconomic and demographic data

The socioeconomic and demographic data comes from *China Statistical Yearbook 2000–2020* [44], including economic density, population density, crop acreage, output of aquatic products, real estate development investment, population of year-end, proportion of age 0–14, proportion of age 65 and over, proportion of urban population, provincial GDP, per capita GDP, per capita annual disposable income of urban residents, per capita annual disposable income of rural residents, proportion of illiterate (semi-illiterate), listener rating, viewer rating, beds of medical institutions per 10,000 populations, medical technical personnel of per 10,000 populations and licensed doctor personnel of per 10,000 populations, which are used to reflect the vulnerability and resilience of TSSDs.

Taking the provincial-level administrative region as the statistical object, a total of 76 TSSDL samples in the study area are collected and randomly divided into the original training set (60 samples) and test set (16 samples). The virtual samples are generated

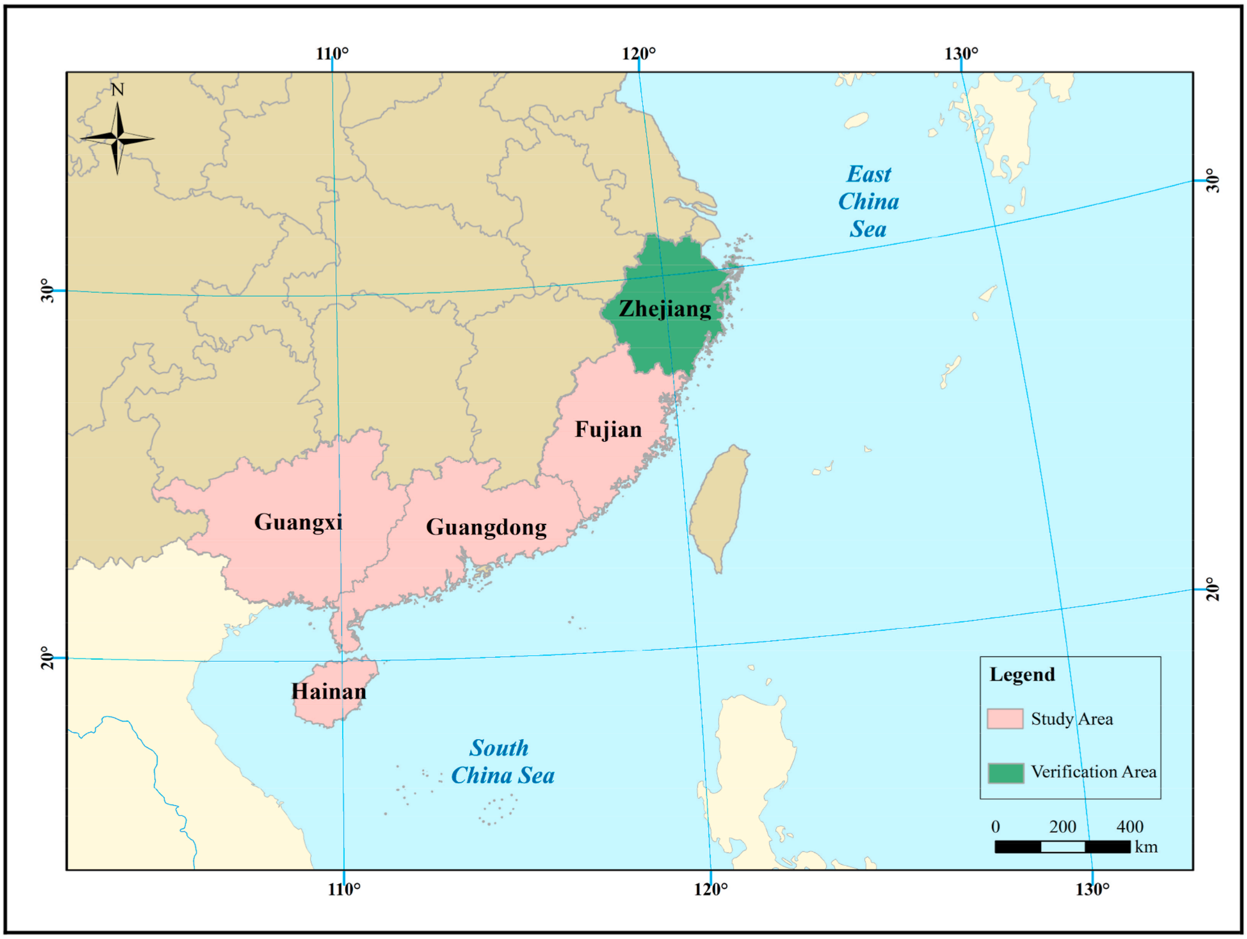

Fig. 2. The study area and the verification area.

with the original training samples. Combining the original training set with the generated virtual samples forms an augmented training set. Besides, the estimation models are trained and optimized based on the augmented training set and the k-fold cross-validation (k-CV) method.

2.2.2. Data normalization

Indicators in the TSSDL indicator system are with different dimensions and orders of magnitude, leading to the inconvenience of direct comparison and numerical analysis between indicators. Besides, in the case of different indicator dimensions, indicators with large values would weaken the influence of indicators with small values. Therefore, data normalization is implemented to improve the rationality of data calculation. According to Eq. (1) [35], this paper adjusts the data to [0,1].

$$x'_{i,j} = \frac{x_{i,j} - x_{\min,j}}{x_{\max,j} - x_{\min,j}} \tag{1}$$

Where i $(i = 1, 2, \cdots, N)$ and j $(j = 1, 2, \cdots, n)$ represent samples' index and indicators' index, respectively; N and n represent the number of samples and indicators, respectively; $x_{i,j}$ and $x'_{i,j}$ represent original data and normalized data of the i_{th} sample corresponding to the j_{th} indicator, respectively; $x_{max,j}$ and $x_{min,j}$ represent the maximum and minimum values of all sample data corresponding to the j_{th} indicator.

2.2.3. Adjustment of economic indicators

Statistical data collected in this paper covers 2000 to 2020, in which the economic indicators are at the current year's price and are called the Nominal economic indicators. However, economic prices over different years have changed dramatically due to inflation. Therefore, economic indicators from other years cannot be compared directly. To eliminate the impact of inflation, the economic indicators in different years need to be adjusted, and the adjusted economic indicators are called the Real economic indicators. This paper uses Eq. (2) to obtain the GDP Deflator α of different years and then uses α and Eq. (3) to adjust the economic indicators [12]. The α for each year from 2000 to 2020 is shown in Fig. 5(a).

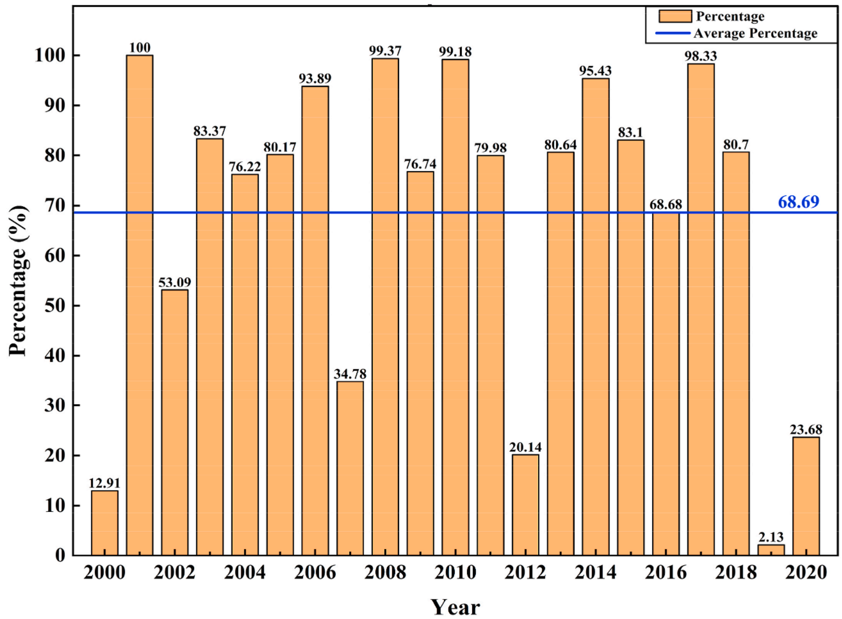

Fig. 3. The percentage of TSSDL in the study area to the nation's (from 2000 to 2020).

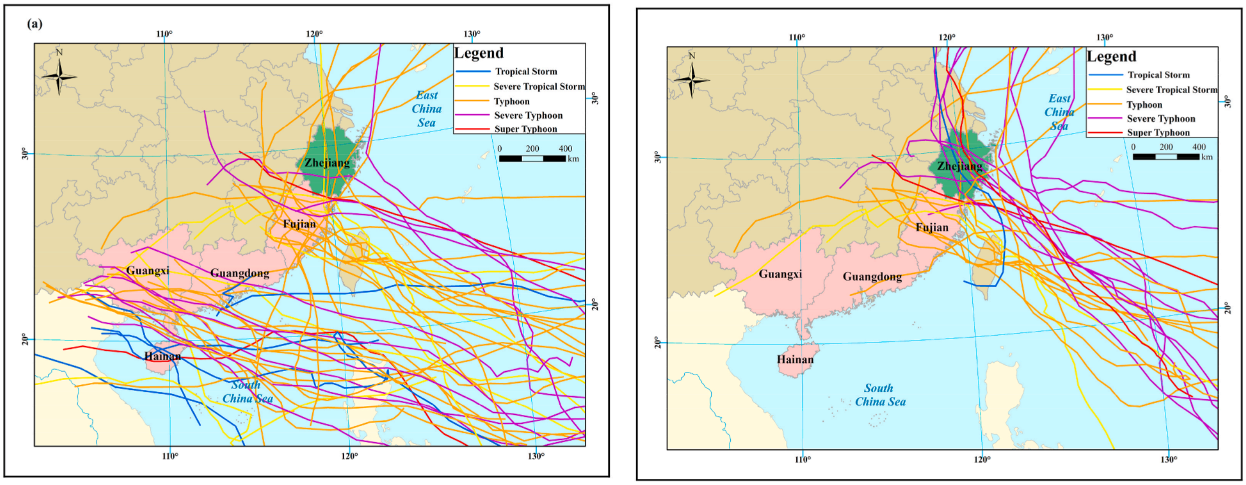

Fig. 4. The typhoon tracks: (a) 52 typhoon tracks in the study area and (b) 26 typhoon tracks in Zhejiang Province.

$$\alpha(m) = \frac{GDP(m) \times \beta(2010)}{GDP(2010) \times \beta(m)} \times 100 \tag{2}$$

$$Real\ \ economic\ indicators\,(m) = Nominal\ economic\ indicators\,(m)\,/\,\alpha\,(m) \tag{3}$$

In Eq. (2), m represents the year that the storm surge disaster event occurred. $\alpha(m)$ represents the GDP deflator corresponding to m year (2010 as the base year, i.e., 100, shown in Fig. 5 (a)). $\alpha(m)$ represents the index of GDP corresponding to m year (1978 as the base year, i.e., 100, obtained from *China Statistical Yearbook* [44], shown in Fig. 5(b)). *Real economic indicators (m)* and *Nominal economic indicators (m)* represent the actual and nominal economic indicators of TSSDs corresponding to m year, respectively.

2.2.4. Grade division standard of TSSDL

The direct economic losses of TSSDs in the same range are classified as the same loss grade, which is in preparation for calculating the indicator's importance to the loss. The grades of TSSDL are divided into four grades according to the direct losses caused by TSSDs. The grade division standard of TSSDL is shown in Table 1 [12,48].

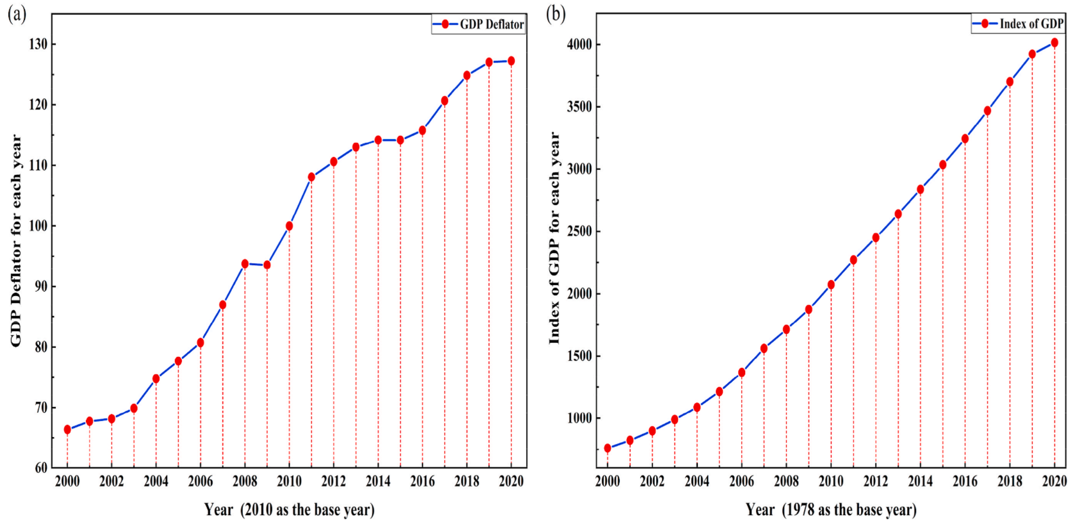

Fig. 5. The economic parameters for years from 2000 to 2020: (a) GDP Deflator and (b) Index of GDP.

Table 1
The grade division standard of TSSDL.

Grades	I (Catastrophe)	II (Disaster)	III (Medium Disaster)	IV (Small Disaster)
Direct economic loss (million RMB)	>5000	2000–5000	1000–2000	<1000

2.3. Construction of indicator system

A comprehensive, reasonable, and objective indicator system is a basis for constructing an accurate estimation model of TSSDL. In the current research, indicator systems are generally built from three aspects: the hazard of disaster-causing factors, the vulnerability of disaster-bearing bodies, and resilience [12,49,50]. This paper follows the successful strategy of pre-researches to construct the indicator system from the three aspects, covering hazard (5 indicators), vulnerability (18 indicators), and resilience (6 indicators), with 29 indicators. These indicators reflect the danger degree of TSSDs, the sensitivity degree of disaster bodies [51], and the disaster mitigation capability of invested resources [52].

2.4. Indicator screening

Because the principle of comprehensiveness was applied to establish the TSSDL estimation indicator system, as many indicators were collected as possible. However, there is redundancy among the indicators, so screening and removing the less significant indicators is essential [12]. Indicator screening reduces the complexity and improves the accuracy of the estimation model, which is a crucial content in machine learning research. In addition, while ensuring the accuracy of the loss estimation model, the number of indicators is smaller after indicator screening, which saves a lot of time and economic cost for collecting indicators data.

The RF-RFECV method, consisting of the random forest-recursive feature elimination (RF-RFE) method and the k-CV method, is applied to screen the indicators for obtaining the optimal subset of indicators (SoI). The principle of the RF-RFECV method [53,54] is shown in Fig. 6. The RF-RFECV contains two phases: in phase 1, the importance of indicators is calculated by the RF-RFE method, and the list of indicator importance ordering is obtained; in phase 2, the optimal SoI is determined by the k-CV method and the ordering list.

The details about the RF-RFE and k-CV methods are introduced below.

2.4.1. The RF-RFE method

The indicator screening is necessary to obtain the indicators with high importance from the TSSDL indicator system. Among the many indicator screening methods, the RF-RFE method is widely used in indicators screening and achieves excellent performance with high classification accuracy and low computational cost [55]. There are four steps to achieve the RF-RFE method. Step 1: the Gini coefficient is calculated by the RF algorithm (Eq. (4)) [56]and taken as the indicator's importance. Step 2: sort the indicators according to their Gini coefficient and obtain an SoI by deleting the least important indicator in the original indicator system. Step 3: Step 1 and Step 2 are repeated recursively until the original indicator system is empty. Step 4: The sorting and screening of all indicators are completed.

$$Gini(t) = \sum_{j=1}^{k} p_{(j/t)} * \left(1 - p_{(j/t)}\right) = 1 - \sum_{j=1}^{k} p_{(j/t)}^{2} \tag{4}$$

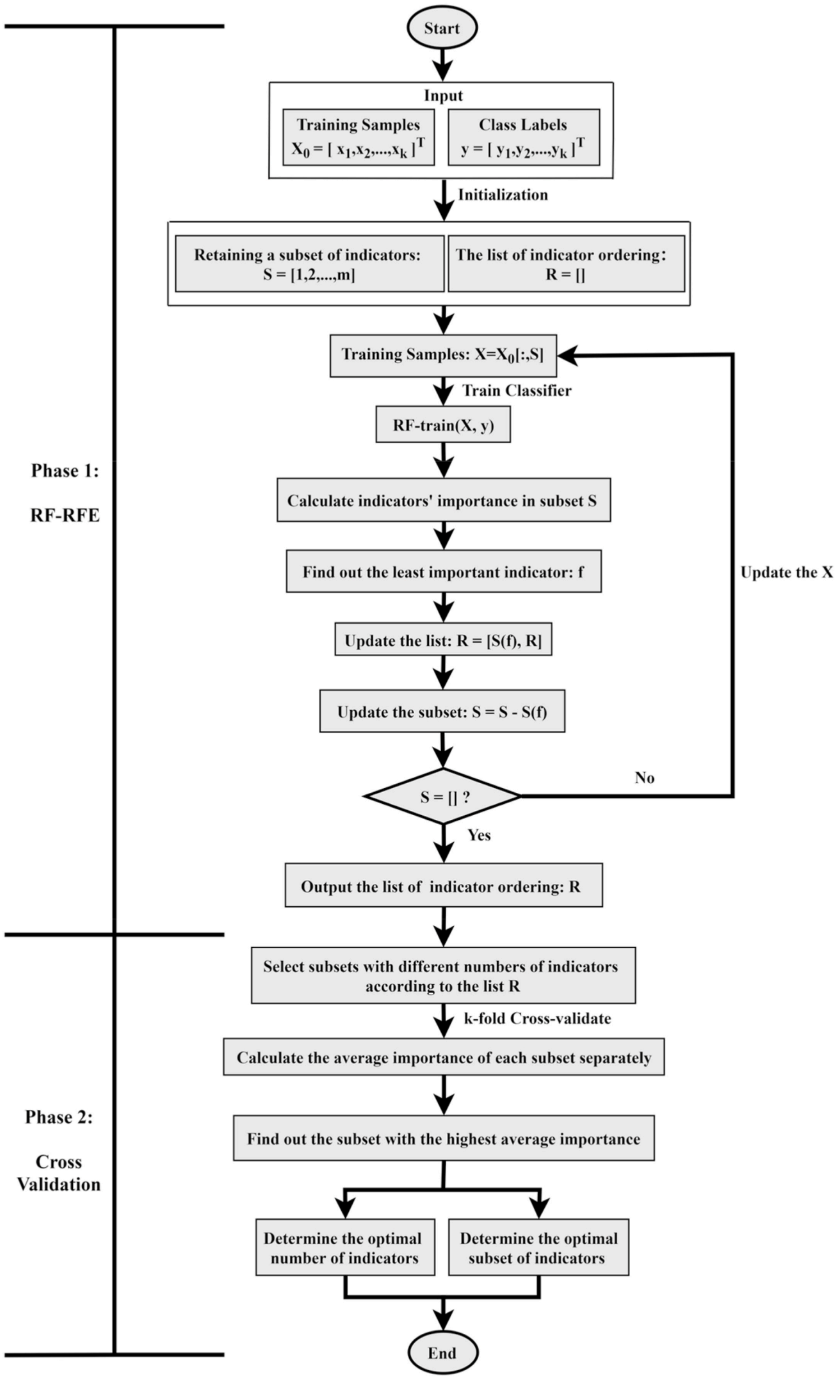

Fig. 6. The flowchart of the RF-RFECV indicator screening method.

In Eq. (4), Gini(t) is the Gini coefficient of node t, $p_{(j/t)}$ represents the occurrence probability of TSSDL grades at node t. The k represents the number of TSSDL grades. The classification standard of TSSDL is shown in Table 1.

2.4.2. The k-CV method

The strong robustness of the loss estimation model is crucial for practical applications, but the typhoon storm surge disaster samples with small data sizes limit the parameter optimization of the loss estimation models, resulting in models prone to overfitting [12]. A scientific dataset partitioning strategy, i.e., the k-fold cross validation method, namely the k-CV method, can improve the generalization ability of the model [57]. The principle of dividing the original training set by the k-CV method is shown in Fig. 7. The k-CV method randomly divides the data set into k equal-sized subsets and takes k-1 subsets as the training set and the remaining one as

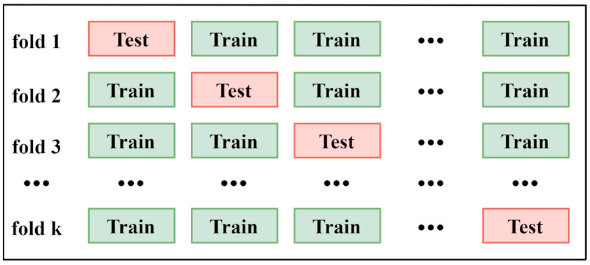

Fig. 7. The schematic diagram of indicators division of the k-CV method.

the test set. Cross-validation is repeated k times until each subset is validated once, and the average value of all validations is taken as the final result. The k-CV method is applied to divide the original training set and calculate the average importance of subsets that contain different numbers of indicators. The optimal SoI is determined by the highest average importance.

2.5. Virtual sample generation

Due to the limitations of small size and incomplete information of TSSDL samples, the small sample problem inevitably occurs when MLMs are applied to TSSDL estimation, i.e., easy over-fitting and poor generalization ability [13]. Virtual sample generation (VSG) is one of the effective methods to solve the small sample problem. The VSG methods are mainly divided into two categories according to different VSG ideas. One is to generate virtual samples by perturbing the original samples, and the other is to generate virtual samples by extracting the nontrivial prior knowledge from the original samples [58].

The VSG method, which combines Gaussian Noise with the Information Diffusion Model based on the Vibrating String equation (named GN-VSIDM), is proposed to generate virtual TSSDL samples. The GN-VSIDM combines the ideas of data perturbation and prior knowledge extraction. As shown in Fig. 8, generating the virtual TSSDL sample by the GN-VSIDM method is divided into four steps. Step 1: Gaussian noises, quantitatively calculated by the hyperparameter SNR, are added to the original TSSDL indicators for data perturbation and generate the virtual indicators. Step 2: the IDM based on the vibrating string equation (VSIDM) extracts the prior knowledge from the original TSSDL samples. Step 3: the virtual TSSDL data is generated by the virtual indicators (from Step 1) and prior knowledge (from Step 2) through approximate reasoning in the VSIDM; Step 4: combining the virtual indicators and virtual TSSDL data generates complete virtual TSSDL samples.

Suppose that the original TSSDL sample set contains N samples and n indicators. The original indicator data is denoted as x_{ij} ($i = 1,2,\cdots,N; j = 1,2,\cdots,n$), the original indicators of the i_{th} sample are denoted as $X_i = (x_{i1},\cdots,x_{in})$, the sample data set of the j_{th} indicator is denoted as $X_j = \{x_{1j},\cdots,x_{Nj}\}$, and the TSSDL data is denoted as $Y = \{y_1,y_2,\cdots,y_N\}$. As shown in Eq. (5) [59], all the TSSDL samples are denoted as S.

$$S = \{s_1,\cdots,s_N\} = \{(x_{11},\cdots,x_{1n},y_1),\cdots,(x_{N1},x_{N2},\cdots,x_{Nn},y_N)\} \tag{5}$$

The formulaic representations of the GN-VSIDM method for generating virtual TSSDL samples are introduced as follows.

2.5.1. Virtual indicators generation

Perturbing the original samples is an important idea to generate virtual samples [58]. The virtual indicators are generated by adding Gaussian noise to perturb the original indicators in this paper [60]. To calculate the suitable Gaussian noise, the hyperparameter SNR [61] is introduced. According to the theory of noise calculation in digital signal processing [61], each original indicator is regarded as a signal, then the power of Gaussian noise can be calculated by the signal power and the SNR.

(1) Average signal power

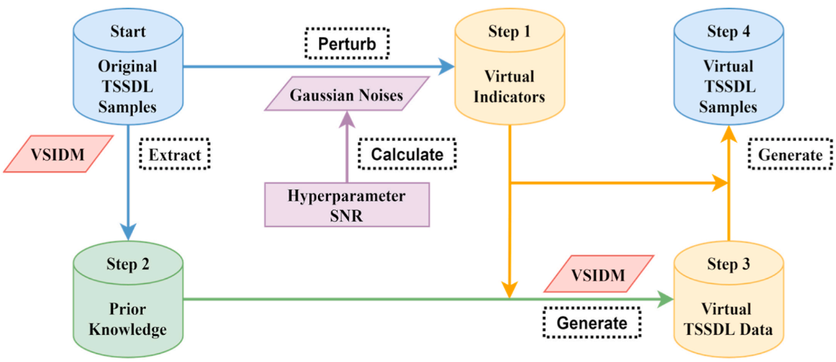

Fig. 8. The schematic diagram of virtual TSSDL sample generation by the GN-VSIDM method.

Each signal is denoted as $X_j = \{x_{1j},\cdots,x_{Nj}\},j = 1,2,\cdots,n$. For the original TSSDL sample set, the number of samples and indicators is limited, and the indicator data is not periodic. Therefore, each signal is a finite-length and aperiodic signal, and the j_{th} signal's average power is represented by Eq. (6) [61].

$$P_s(j) = \left(\sum_{i=1}^{N} \left|x_{ij}\right|^2\right)/\mathrm{N}, j = 1, 2, \cdots, n. \tag{6}$$

(2) Average noise power

The calculation of the average noise power of the j_{th} signal is shown in Eq. (7) [61].

$$P_n\left(j\right) = P_s(j)/(10^{\frac{SNR}{10}}), j = 1,2,\cdots,n. \tag{7}$$

Where SNR is the hyperparameter whose unit is dB.

(3) Gaussian noise value

The numerical value of the Gaussian noise is calculated by Eq. (8) [60,61].

$$n_{ij} = \sqrt{P_n(j)} \times \alpha_{ij}, j = 1, 2, \cdots, n. \tag{8}$$

Where n_{ij} represents the Gaussian noise value of the j_{th} indicator in the i_{th} sample. α_{ij} is a random number, obeying the standard normal Gaussian distribution (mean 0 and standard deviation 1, i.e., $\alpha_{ij} \in N(0,1)$).

(4) The virtual indicators

As shown in Eq. (9) [60,61], the virtual indicators data is generated by the original indicator data and Gaussian noise values.

$$x_{v,ij} = x_{ij} + n_{ij}, j = 1, 2, \cdots, n; i = 1, 2, \cdots, N. \tag{9}$$

Where v represents the virtual generation.

The virtual indicators, generated by the original indicators of the i_{th} TSSDL sample $(x_{i1},\cdots,x_{in})$ and the Gaussian noise perturbation (a part of the GN-VSIDM), are denoted as $(x_{v,i1},\cdots,x_{v,in})$.

This subsection introduces the generation of virtual indicators, i.e., Step 1 of the GN-VSIDM in Fig. 8.

2.5.2. Prior knowledge extraction

Extracting the nontrivial prior knowledge from the original samples is another important idea to generate virtual samples. The TSSDL sample data is highly nonlinear, while the VSIDM extracts more prior knowledge from the nonlinear data and processes the abnormal distribution information well [35,42]. Therefore, the VSIDM is applied to extract the prior knowledge form the original TSSDL samples. The following is a detailed introduction to the prior knowledge extraction in the VSIDM.

(1) Build the universe of discourse U of the information diffusion

Let U_j represent the universe of discourse to diffuse indicator X_j, and U_{j+1} be used to diffuse the TSSDL data Y. Let $\lambda = n + 1$, the two universes of indicators and TSSDL are combined to form a α-dimensional universe of discourse. The combined α-dimensional universe is shown in Eq. (10) [59].

$$U_1 \cdots \times U_j \cdots \times U_\lambda \tag{10}$$

Where $U_j = \{u_{j1},u_{j2},\cdots,u_{jm}\},j = 1,2,\cdots,\lambda$; m is the number of monitoring points in the universe of discourse.

(2) Build the α-dimensional information diffusion equation

The α-dimensional information diffusion equation is obtained by combining the vibrating string equation and the normal information diffusion equation. Let $y_i = x_{i\lambda}, i = 1,2,\cdots N$; then, the i_{th} sample $s_i = (x_{i1},x_{i2},\cdots,x_{in},y_i)$ of S (Eq. (5)) is denoted as a α-dimensional sample, i.e. x_i, and the expression is shown in Eq. (11) [59].

$$x_i = (x_{i1}, x_{i2}, \cdots, x_{i\lambda}) \in S \tag{11}$$

A α-dimensional monitoring point in the universe of discourse (Eq. (10)) is denoted as Eq. (12) [59].

$$u = \left(u_{1k_1}, \cdots, u_{jk_j}, \cdots, u_{\lambda k_\lambda}\right) \in U_1 \times U_2 \cdots \times U_\lambda \tag{12}$$

Where $k_j \in \{1,2,\cdots,m\}, j = 1,2,\cdots,\lambda$.

As is shown in Eq. (13), the α-dimensional information diffusion equation [42,59] is used to spread the information of x_i to the α-dimensional monitoring point u. This formula calculates the amount of information spread to each monitoring point for the input indicator data in a single sample.

$$\mu(x_i, u) = \prod_{j=1}^{\lambda} \exp\left(-\left|x_{ij} - u_{jk_j}\right| / d_j\right) / 2 \tag{13}$$

Where d represents the diffusion coefficient of the vibrating string diffusion equation.

(3) Calculate the diffusion coefficient d

Suppose the diffusion coefficient of the indicator X_j on the universe of discourse is denoted as d_j, which can be calculated by Eq. (14) [42].

$$d_j = \mathrm{k} * \left(\max\left(X_j\right) - \min\left(X_j\right)\right) / (N-1), j = 1, 2, \cdots, \lambda. \tag{14}$$

Where k is a constant related to the number of samples N. The specific corresponding relationship is shown in Table 2 [42].

(4) Calculate the information matrix Q

Let the TSSDL samples' diffusion information at the α-dimensional monitoring point u be $Q_{k_1k_2\cdots k_\lambda}$, and its expression is shown in Eq. (15) [59]. This formula calculates the total amount of information diffused by the input data of all disaster samples at each monitoring point, also as the information matrix calculation.

$$Q_{k_1k_2\cdots k_\lambda} = \sum_{i=1}^{N} \mu(x_i, u) \tag{15}$$

Substituting Eq. (13) into Eq. (15), the specific form of $Q_{k_1k_2\cdots k_\lambda}$ is obtained, as shown in Eq. (16) [42,59].

$$Q_{k_1k_2\cdots k_\lambda} = \sum_{i=1}^{N} \prod_{j=1}^{\lambda} \exp\left(-\left|x_{ij} - u_{jk_j}\right| / d_j\right) / 2 \tag{16}$$

According to Eq. (16), the information matrix Q is calculated, representing the information accumulation of TSSDL samples S on the α-dimensional universe of discourse U, as shown in Eq. (17) [59].

$$Q = \left\{Q_{k_1\cdots k_j\cdots k_\lambda}\right\}_{\underbrace{\mathrm{m} \times \cdots \times \mathrm{m} \cdots \times \mathrm{m}}_{\lambda}} \tag{17}$$

Where $k_j \in \{1, 2, \cdots, m\}, j = 1, 2, \cdots, \lambda$.

(5) Calculate the fuzzy relationship matrix R

The fuzzy relationship matrix R can be calculated by the information matrix Q, as shown in Eq. (18) and Eq. (19) [59].

$$S_{k_\lambda} = \max_{\substack{1 \le k_j \le m \\ 1 \le j \le \lambda - 1}} \left\{Q_{k_1\cdots k_j\cdots k_\lambda}\right\} \tag{18}$$

In Eq. (18), S_{k_λ} is the maximum amount of information on the $U_1 \times U_2 \cdots \times U_{\lambda-1}$ space corresponding to each monitoring point in the λ_{th} dimensional space.

$$r_{k_1k_2\cdots k_{\lambda-1}k_\lambda} = Q_{k_1k_2\cdots k_{\lambda-1}k_\lambda} / S_{k_\lambda} \tag{19}$$

Where $r_{k_1k_2\cdots k_{\lambda-1}k_\lambda}$ is the fuzzy information at the monitoring point u.

Then, the fuzzy relationship matrix R is obtained by Eq. (20) [59].

Table 2
The values of the k-parameter in VSIDM.

N	2	3	4	5	6	7
k	1.75260	1.55885	1.49338	1.46630	1.45405	1.44826
N	8	9	10	11	12	>12
k	1.44544	1.44406	1.44338	1.44303	1.44270	1.44270

$$R = \left\{ r_{k_1 k_2 \cdots k_{\lambda-1} k_\lambda} \right\}_{m_1 \times m_2 \times \cdots \times m_\lambda} \tag{20}$$

The prior knowledge of the TSSDL samples, i.e., fuzzy relationship matrix R, is extracted through the information diffusion equation in the VSIDM. The matrix R reflects the fuzzy causal relationship between indicators and TSSDL data.

This subsection achieves the prior knowledge extraction, i.e., Step 2 of the GN-VSIDM in Fig. 8.

2.5.3. Virtual TSSDL generation

(1) Information Distribution

Since fuzzy inference based on fuzzy matrices is required, fuzzification of the input data is necessary [35]. The virtual indicators are turned into a fuzzy set to generate the virtual TSSDL data with the approximate reasoning method.

Let the virtual indicators of an original TSSDL sample be $X_v = (x_{v,1}, x_{v,2}, \cdots, x_{v,\lambda-1})$ and the $\lambda - 1$ dimensional monitoring point in the universe of discourse be $u_{\lambda-1} = (u_{1k_1}, u_{2k_2}, \cdots, u_{\lambda-1 k_{\lambda-1}}) \in U_1 \times U_2 \cdots \times U_{\lambda-1}, k_\lambda = 1,2,\cdots,m$. This paper uses the $\lambda - 1$ dimensional linear information distribution equation (as shown in Eq. (21) and Eq. (22) [59] to convert X_v to a fuzzy set on the universe of discourse $U_1 \times U_2 \cdots U_{\lambda-1}$. The fuzzy set is denoted as $\widetilde{A}$.

$$q_{k_1 k_2 \cdots k_{\lambda-1}} = \prod_{j=1}^{\lambda-1} f\left(X_{v,j}, u_{jk_j}\right) \tag{21}$$

$$f\left(X_{v,j}, u_{jk_j}\right) = \begin{cases} 1 - \dfrac{\left|X_{v,j} - u_{jk_j}\right|}{u_{j2} - u_{j1}}, & \left|X_{v,j} - u_{jk_j}\right| < u_{j2} - u_{j1} \\ 0, \text{otherwise} & \end{cases} \tag{22}$$

Where $q_{k_1 k_2 \cdots k_{\lambda-1}}$ represents the degree of membership. $k_j \in \{1,2,\cdots,m\}, j = 1,2,\cdots,\lambda - 1$.

(2) Normalize the degree of membership

As shown in Eq. (23) [59], all degrees of membership of the fuzzy set $\widetilde{A}$ are normalized by the maximum value of the degrees of membership.

$$\begin{cases} a_{k_1 \cdots k_j \cdots k_{\lambda-1}} = q_{k_1 k_2 \cdots k_{\lambda-1}} / s \\ s = \max\limits_{\substack{1 \le k_j \le m \\ 1 \le j \le \lambda-1}} \left\{ q_{k_1 k_2 \cdots k_j \cdots k_{\lambda-1}} \right\} \end{cases} \tag{23}$$

Where $a_{k_1 \cdots k_j \cdots k_{\lambda-1}}$ represents the normalized degree of membership. $k_j \in \{1,2,\cdots,m\}, j = 1,2,\cdots,\lambda - 1$.

(3) Fuzzy composition

The $\widetilde{B}$ with the membership function $\mu_B(u_{\lambda k_\lambda})$ is the fuzzy output about the fuzzy input $\widetilde{A}$. According to Eq. (24) [59], the fuzzy set $\widetilde{B}$ is calculated by the 'max-min' composition rule. Besides, the normalized degree of membership and the fuzzy information R (i.e., the prior knowledge extracted from original TSSDL samples) are used to calculate the membership function.

$$\mu_B\left(u_{\lambda k_\lambda}\right) = \max_{\substack{1 \le k_j \le m \\ 1 \le j \le \lambda-1}} \min \left\{ a_{k_1 k_2 \cdots k_j \cdots k_{\lambda-1}}, r_{k_1 k_2 \cdots k_j \cdots k_{\lambda-1} k_\lambda} \right\} \tag{24}$$

(4) Defuzzification

Obtained by approximate inference and fuzzy input data, the loss is fuzzy data and defuzzification is necessary to obtain a crisp loss value. The Center of Gravity (COG) method is used for defuzzification to map the fuzzy output to a crisp estimated TSSDL value $\widetilde{y}$, as shown in Eq. (25) [59].

$$\widetilde{y} = \frac{\sum\limits_{k_\lambda=1}^{m} \mu_B\left(u_{\lambda k_\lambda}\right) * u_{\lambda k_\lambda}}{\sum\limits_{k_\lambda=1}^{m} \mu_B\left(u_{\lambda k_\lambda}\right)} \tag{25}$$

The virtual TSSDL is a generation of the virtual indicators data, the prior knowledge (i.e., the fuzzy relationship matrix R), and the approximate reasoning method of the GN-VSIDM.

This subsection achieves the virtual TSSDL data generation, i.e., Step 3 of the GN-VSIDM in Fig. 8.

2.5.4. Generation of virtual TSSDL samples

As shown in Eq. (26) [62], the virtual indicators ($x_{v,i1}$,⋯,$x_{v,in}$) and the virtual TSSDL $\tilde{y}_i$, generated by the i_{th} TSSDL sample and the GN-VSIDM method, are combined to generate the completed virtual samples.

$$S_{v,i} = \left(X_{v,i}, \tilde{y}_i\right) = \left(x_{v,i1}, x_{v,i2}, \cdots, x_{v,in}, \tilde{y}_i\right), i = 1, 2, \cdots, N \tag{26}$$

The complete virtual TSSDL samples are generated, i.e., Step 4 of the GN-VSIDM in Fig. 8.

2.5.5. The common virtual sample generation methods

The performance of the GN-VSIDM is verified by comparing it with common VSG methods. Two traditional VSG methods, namely Gaussian Noise Injection (GNI) and the Bootstrap methods, are selected for comparison [13]. The basic principles of the two ways are shown as follows.

(1) Gaussian Noise Injection (GNI)

The strategy of the GNI method is to add Gaussian noise to the original indicators for perturbing the original input data, while the output data remains unchanged [13]. The specific operation is shown in Eq. (27) and Eq. (28) [13].

$$X_{GNI,i} = X_i + \vec{\alpha} \circ X_i \tag{27}$$

$$y_{GNI,i} = y_i \tag{28}$$

Where $\vec{\alpha} = \left(\alpha_1, \alpha_2, \cdots, \alpha_n\right)$ obey the normal distribution with a mean 0 and standard deviation α. The symbol '∘' is the Hadamard product operation, representing the operation of multiplying the corresponding elements of two matrices of the same dimensions.

(2) Bootstrap

Different from the GN-VSIDM and GNI methods, Bootstrap's strategy is random sampling with replacement [63]. The basic principle of the Bootstrap method to generate virtual samples is to create a series of random integers i (i_1,i_2,⋯,i_N∈[1,N]) randomly, where N is the size of the original TSSDL sample set. Then, these random integers are used as sample indices to sample the original TSSDL sample set with the resampling strategy. The sampling is repeated K times, and the number of sampled samples is N*K.

2.6. Machine learning models (MLMs)

This paper uses three machine learning methods, LSBoost, XGBoost, and RF, to estimate TSSDL. The following is a brief introduction to these three methods.

2.6.1. The least-squares boosting (LSBoost)

The LSBoost model, proposed by Friedman [64], is a gradient boosting method that uses the least-squares method as the loss function to minimize the mean squared error. The LSBoost model provides excellent fitting results and becomes one of the best regression methods.

2.6.2. eXtreme gradient boosting (XGBoost)

The XGBoost model is an excellent ensemble learning method, which improves the model's robustness by using regularization and improves the running speed by parallelization strategy. The XGBoost model uses cross-validation in the boosting iteration process and obtains better performance under the condition of small samples [65].

2.6.3. Random forest (RF)

The RF method is a classic ensemble learning algorithm with excellent fitting ability. Characterized by high robustness and insensitivity to parameter settings, the estimation results of the RF method are reliable and easy to understand [66]. Besides, the RF method can also calculate and order the importance of indicators.

2.7. Model evaluation methods

In this paper, root mean squared error (RMSE) and coefficient of determination (R^2) are used for model evaluation.

2.7.1. Root mean squared error (RMSE)

The RMSE measures the deviation between the observed and actual values, calculated by Eq. (29) [62]. The smaller the RMSE, the higher the model performance.

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{N}\left(y_i - \hat{y}_i\right)^2}{N}} \tag{29}$$

where N is the number of samples.

The EIR_RMSE is the error improvement rate of RMSE, which is calculated from the changed value of RMSE before and after model optimization, calculated by Eq. (30) [62]. The EIR_RMSE reflects the change of the model's RMSE. A positive value of EIR_RMSE indicates that the model performance is improved, and the greater the value, the greater the improvement. Conversely, a negative value of EIR_RMSE indicates a degraded model performance.

$$EIR\_RMSE = \frac{RMSE_{before} - RMSE_{after}}{RMSE_{before}} \times 100\% \tag{30}$$

2.7.2. Coefficient of determination (R^2)

The calculation of R^2 is shown in Eq. (31) and Eq. (32) [67]. The model's fitting ability is better when the value of R^2 is closer to 1.

$$\bar{y}_i = \frac{\sum_{i=1}^{N} y_i}{N} \tag{31}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{N} (y_i - \bar{y}_i)^2} \tag{32}$$

Where y_i represents the actual value, $\bar{y}_i$ represents the statistical mean of all actual values, and $\widetilde{y}_i$ represents the fitted value obtained by the estimation model.

2.8. Model robustness

MLMs are required to perform well not only on the test set but also on a new data set. Robustness is a significant evaluation metric for MLMs, reflecting the model's stability on new datasets. It is a widely used method to build an optimal model using a training dataset (in the study area) and verify its robustness with a validation set (in another region) [13,34]. In this paper, after constructing and evaluating the optimal TSSDL estimation model with 76 samples in the study area, the robustness of the optimal model is verified with 26 samples in Zhejiang Province.

3. Results and analysis

3.1. The results of indicator system construction

The TSSDL estimation indicator system, which includes 29 indicators, is constructed from three aspects: the hazard of disaster-causing factors (5 indicators), the vulnerability of disaster-bearing bodies (18 indicators), and resilience (6 indicators). The information on indicators is shown in Table 3.

3.2. Screening of indicators

Indicator screening aims to obtain an optimal subset of indicators (SoI) whose indicators are characterized by a small number and high cumulative importance. As shown in Fig. 9 (a), when the indicators' number is seven, the cumulative importance reaches 0.6739 of the whole. That means when the indicator's number accounts for a quarter of the indicator system, and the importance accounts for two-thirds importance of the indicator system. Therefore, seven is the optimal indicator number of the SoI.

The RF-RFECV method was used to determine the specific indicators of the SoI by ordering the importance of the indicators. The RF-RFECV method was executed 20 times, and seven indicators with the highest rank of importance were selected and recorded in each experiment. The number of times each indicator was selected as the top seven important indicators is shown in Fig. 9 (b). In Fig. 9 (b), the top seven indicators that form the optimal SoI are X1 (Maximum water increase), X6 (Affected population), X7 (Disaster area of mariculture), X8 (Submerged farmland area), X21 (Per capita annual disposable income of rural residents), X22 (Coastal facilities), and X23 (Damaged fishing boats). Among them, X1 belongs to the hazard of disaster-causing factors; X6, X7, X8, X21, X22, and X23 belong to the vulnerability of disaster-bearing bodies.

Among the indicators of the optimal SoI, except for X21 (selected 16 times), the remaining indicators were selected in every experiment (20 times), indicating that the indicators' importance of the optimal SoI is significantly higher than other indicators. Furthermore, the importance of the hazard indicators and the vulnerability indicators are much greater than the resilience indicators, which meet the widely recognized theory that Loss (or risk) = Hazard × Vulnerability [12,68].

In conclusion, the RF-RFECV method reduces the number of indicators and ensures the high cumulative importance of the SoI. Meanwhile, the SoI is interpretable and consistent with the widely accepted theory. The above indicates that the RF-RFECV method is objective, reasonable, and practical for indicator screening.

3.3. Generation of virtual sample

3.3.1. The optimal value of SNR

To determine the optimal value of hyperparameter SNR in the GN-VSIDM method, experiments were carried out with the SNR values ranging from 10 dB to 90 dB (with an interval of 10 dB). The adoption of 5-fold cross-validation ensured the objectiveness and accuracy of experiment results. To exclude the influence of different augmentation multiples (AMs) and ensure high computational effi-

Table 3
The estimation indicator system of TSSDL.

Category	Indicator	Short name	Unit	Data source
Hazard	Maximum water increase	X1	m	Collection of Storm Surge Disasters Historical Data in China 1949–2009 [45]
	Ultra-warning water level	X2	m	Bulletin of China Marine Disaster (https://www.mnr.gov.cn/sj/sjfw/hy/gbgg/zghyzhgb/) [7]
	Maximum wind speed	X3	m/s	National Meteorological Centre Typhoon Website (http://typhoon.nmc.cn/web.html)
	Central pressure	X4	hPa	Zhejiang Province Ocean Forecasting Website (https://zjocean.org.cn/typhoon)
	Forward speed	X5	km/h	
Vulnerability	Affected population	X6	10,000 person	China Statistical Yearbook 2000–2020 [44]
	Disaster area of mariculture	X7	1000 ha	
	Submerged farmland area	X8	1000 ha	
	Crop acreage	X9	10,000 mu	
	Output of aquatic products	X10	10,000 ton	
	Real estate development investment	X11	100 million yuan	
	Population of year-end	X12	10,000 person	
	Proportion of age 0–14	X13	%	
	Proportion of age 65 and over	X14	%	
	Proportion of urban population	X15	%	
	Provincial GDP	X16	100 million yuan	
	Per capita GDP	X17	yuan	
	Population density	X18	persons/km^2	
	Economic density	X19	10,000 yuan/km^2	
	Per capita annual disposable income of urban residents	X20	yuan	
	Per capita annual disposable income of rural residents	X21	yuan	
	Coastal facilities (embankments and berms)	X22	km	
	Damaged fishing boats	X23	ship	
Resilience	Proportion of illiterate (semi-illiterate)	X24	%	China Statistical Yearbook 2000–2020 [44]
	Listener rating	X25	%	
	Viewer rating	X26	%	
	Beds of medical institutions per 10,000 populations	X27	set	
	Medical technical personnel of per 10,000 populations	X28	person	
	Licensed doctor personnel of per 10,000 populations	X29	person	

ciency, the number of generated virtual samples was set to three multiples of the original training samples. In the case of virtual samples generated by different SNR values, the performance of the TSSDL estimation model base on three MLMs, i.e., LSBoost, XGBoost, and RF, is shown in Fig. 10.

As shown in Fig. 10, compared with the RMSE of the original samples-based models, the RMSE of the virtual samples-based models is significantly decreased, and the EIR_RMSE is positive, indicating that the generation of virtual samples improves the performance of the three MLMs. Overall, when the SNR is 40–60 dB, the performance of the virtual samples-based models is higher. Besides, the optimal SNR is 50 dB when the RF model reaches the maximum EIR_RMSE of 29.25%, and the XGBoost model achieves the smallest RMSE of 0.1138.

In summary, the GN-VSIDM method improves the TSSDL estimation performance of the three MLMs, and the optimal SNR for generating virtual samples is in the range of 40–60 dB.

3.3.2. The optimal multiple of VSG

The effects of different AMs on the performance of the three MLMs are shown in Fig. 11. In experiments, the SNR value was set to 50 dB.

As shown in Fig. 11, when the AMs are taken from one to seven (with an interval of one multiple), the performances of three MLMs are improved. Furthermore, the models' performances are enhanced significantly when the AMs are taken four, five, and six. Especially, the optimal performance occurs when the AMs are five, when the RF model peaks at the maximum EIR_RMSE of 33.13%, and the XGBoost model achieves the minimum RMSE of 0.1089.

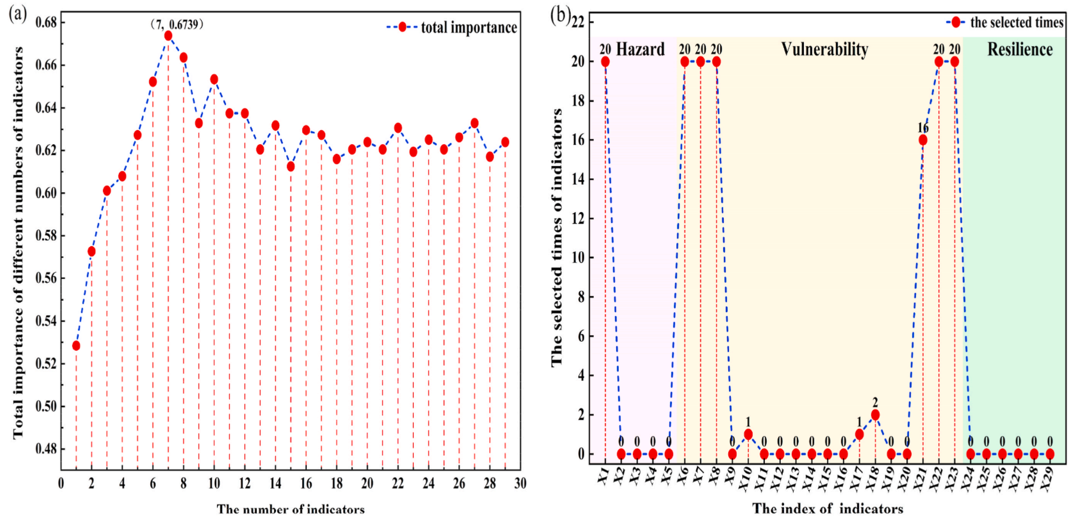

Fig. 9. The indicator screening results by the RF-RFECV method: (a) The corresponding relationship between the number of indicators and their cumulative importance and (b) The number of selected times for each indicator (repeating 20 experiments).

In conclusion, when the SNR is 50 dB, and the range of augmentation multiples is four, five, and six multiples, the GN-VSIDM method makes the three MLMs perform better. Besides, the optimal AMs are five. This conclusion supported the subsequent experimental design.

3.3.3. Comparison of different VSG methods

The effects of different VSG methods, i.e., GN-VSIDM, GNI, and Bootstrap methods, on the three MLMs are shown in Fig. 12 (test set). The SNR of the GN-VSIDM method is set to 50 dB, which is determined according to the experimental results in Section 3.3.1. The standard deviation of Gaussian noise added in the GNI method is 0.05 [13]. The Bootstrap method resamples 60 samples each time.

According to Fig. 12, the following findings are obtained.

(1) The performance of the models based on the original samples is the worst, indicating that the performance of the MLMs is poor under the small sample conditions. However, the models based on the virtual samples generated by the GN-VSIDM method obtain the best performance. Therefore, the GN-VSIDM method is effective and feasible for solving the small sample problem. Besides, the GN-VSIDM method is compared with the GNI and Bootstrap methods to evaluate its performance better.
(2) Compared with the baseline, i.e., the performance of the models based on the original samples, both the GN-VSIDM and the Bootstrap methods reduce the RMSE of the three MLMs under the case of sample augmentation, thus improving the model performance. For the GNI method applied to the XGBoost model, when the AMs are one to six, the GNI method improves the performance of the three MLMs. However, when the AMs reach seven multiples, the GNI method weakens the performance of the three MLMs, even worse than the baseline. The reason for this phenomenon is that the GNI method only adds Gaussian noise to the original indicator data without changing the TSSDL data of the sample. With the increase of AMs, the noise increases, leading to the destruction of the original samples' information distribution and the model performance degradation.
(3) The GN-VSIDM method makes the three MLMs reach the smallest RMSE and perform best when the AMs are five. Therefore, compared with the Bootstrap method, the GN-VSIDM method has a better effect. In fact, the Bootstrap method produces a relatively good result. However, the VSG strategy of the Bootstrap method is based on resampling of the original samples, which is unable to introduce new sample information to fill the information gaps in small samples [69]. Thus, Bootstrap's ability to solve the small sample problem is more limited.
(4) Compared with the GNI method, besides adding Gaussian noise to the original indicator data for perturbation, the GN-VSIDM method generates virtual TSSDL data by approximate reasoning and prior knowledge. The virtual TSSDL contains much new information, making the virtual samples closer to the information distribution of the original dataset. Compared with the Bootstrap method, the GN-VSIDM method obtains new sample information by approximate reasoning and prior knowledge, so this method is better than the Bootstrap method in improving model performance. In summary, the GN-VSIDM method is the most reasonable and practical among the three VSG methods.

3.4. The optimal estimation model

The four data samples, GN-VSIDM-generated, Bootstrap-generated, GNI-generated, and the original training samples, were combined with the three MLMs respectively to obtain 12 joint models. The performance of these 12 joint models was obtained through cross-validation, as shown in Fig. 13. Besides, the main parameters of the three MLMs are shown in Table 4.

In Fig. 13, darker colors represent better performance. Overall, based on the same machine learning algorithm, the model performance based on the virtual samples generated by the GN-VSIDM method is better than Bootstrap-generated, GNI-generated, and the

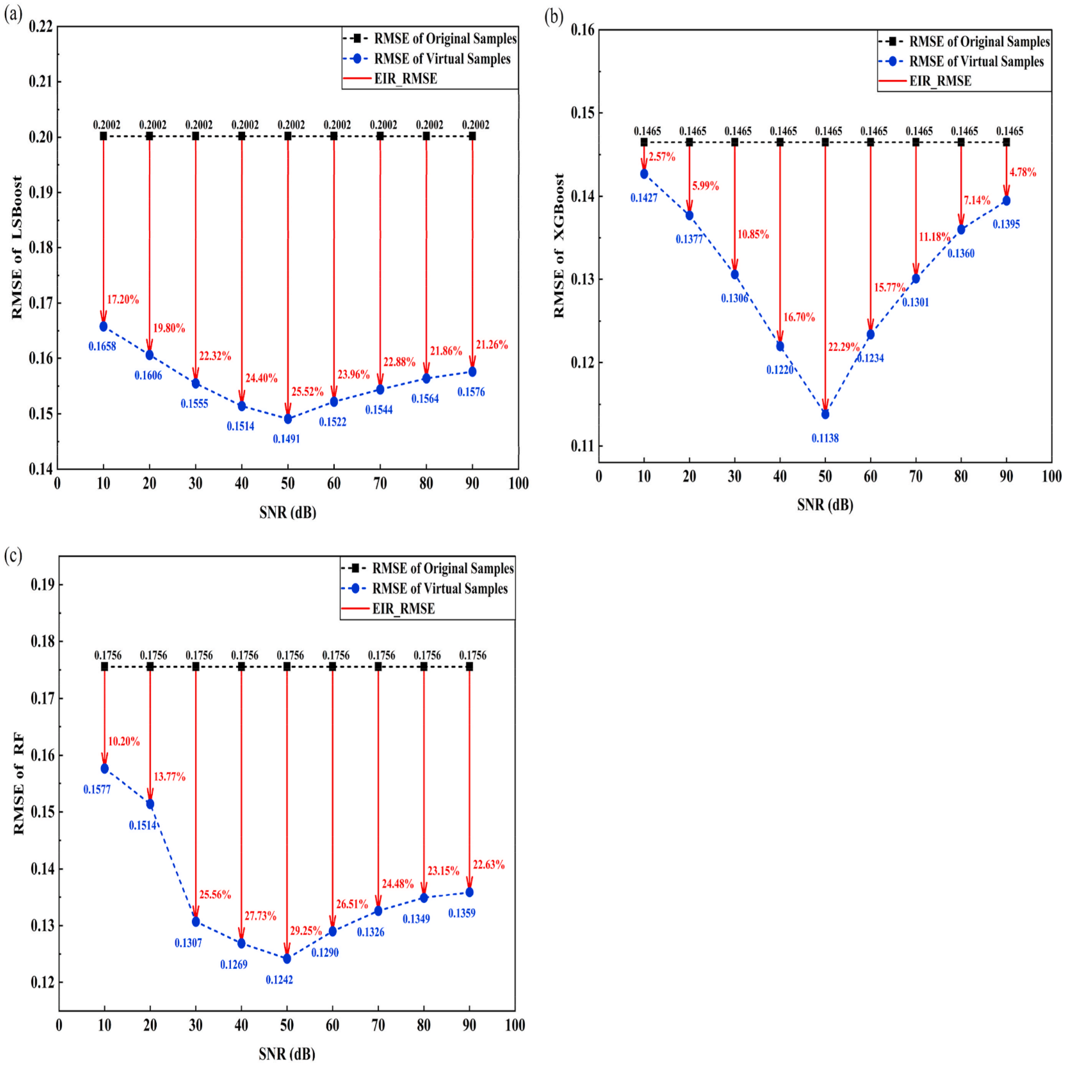

Fig. 10. The effects of different SNR values on the performance of three models: (a) LSBoost model, (b) XGBoost model, and (c) RF model.

original training samples. Besides, based on the same VSG method, the performance obtained by the XGBoost model is the best. Among the 12 joint models, the GN-VSIDM-XGBoost joint model has the smallest RMSE of 0.1089 and the largest R^2 of 0.8292, which reduces 25.67% and improves 19.88%, respectively. Therefore, the GN-VSIDM-XGBoost joint model is the optimal TSSDL estimation model. Besides, the GN-VSIDM method shows an excellent performance in solving the small sample problem of easy overfitting and weak generalization ability.

To further verify the performance of the GN-VSIDM-XGBoost model, it is compared with the performance of three traditional MLMs, including XGBoost, Back Propagation Neutral Network (BPNN), and Decision Tree (DT), and the results are shown in Table 5. The conclusions are as follows: (1) Compared with the XGBoost model, the GN-VSIDM-XGBoost model has a better performance, which means that the GN-VSIDM method improves the poor generalization ability of the MLMs under small sample conditions. (2) The GN-VSIDM-XGBoost model performs significantly better than the BPNN and DT models, and the reason is that the GN-VSIDM-XGBoost model extracts more information from the original small samples and has a better fitting ability and generalization capability. (3) To sum up, the GN-VSIDM-XGBoost outperforms the other three models and is the optimal TSSDL estimation model under small sample conditions.

3.5. Application and verification of the optimal estimation model

3.5.1. The estimation of TSSDL

The GN-VSIDM-XGBoost model is determined to be the optimal joint model through multiple experiments. The parameters of the GN-VSIDM-XGBoost model are shown in Table 4, its SNR is 50 dB, and the augmentation multiples of virtual samples are five. Based on the GN-VSIDM-XGBoost model, the losses of TSSDs (2000–2020, 76 samples) in the study area are estimated, and the results are shown in Fig. 14.

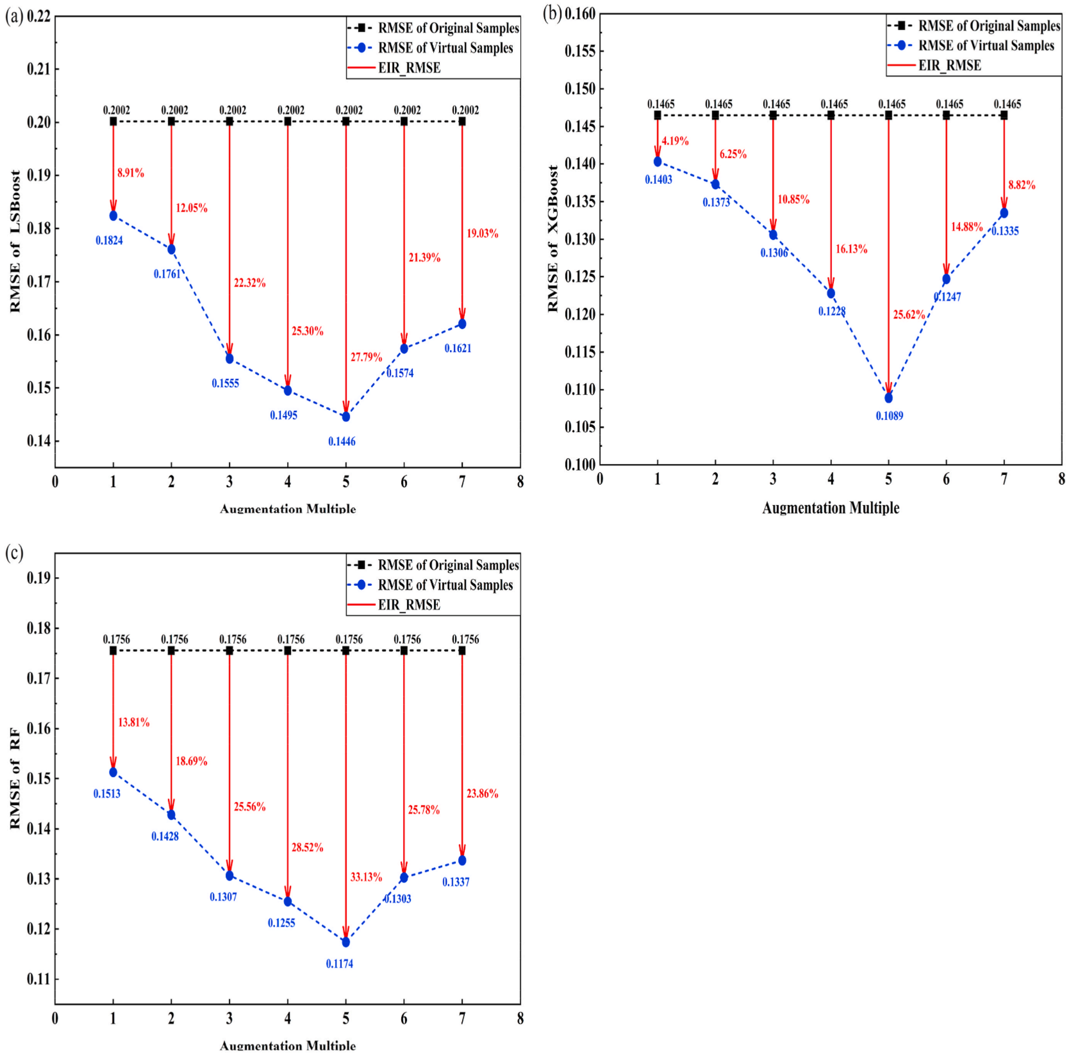

Fig. 11. The effects of different AMs on the performance of different models (test set): (a) LSBoost model, (b) XGBoost model, and (c) RF model.

The samples are divided randomly. For the 60 training samples, the estimation loss curve is similar to the actual loss curve, indicating that the GN-VSIDM-XGBoost model learns the information of the training samples fully and obtains an excellent ability to fit the TSSDL. For the 16 test samples, the estimated loss curve fits well with the actual loss curve, except for the 66th sample with a relatively larger error. In conclusion, the GN-VSIDM-XGBoost model has an excellent fitting ability on the training set and a good generalization ability on the test set.

3.5.2. Robustness verification

The 26 TSSDL samples in Zhejiang Province (2000–2020) were used to verify the robustness of the GN-VSIDM-XGBoost model. Divided these samples into 20 training samples and 6 test samples randomly, and the TSSDL estimation was carried out based on the GN-VSIDM-XGBoost model.

As shown in Fig. 15, the estimated loss curve fits the actual loss curve well, indicating that the GN-VSIDM-XGBoost model is robust because it has excellent performance on a new dataset.

4. Discussion

Based on the GN-VSIDM VSG method and MLM, a high-accuracy and strong robustness TSSDL estimation model is obtained under small sample conditions. The high accuracy ensures that the model fully learns the prior knowledge of historical data, and the strong robustness ensures that the model has a good ability to estimate new disaster data. Therefore, the high-accuracy loss estimation model can provide accurate and effective data support for disaster prevention and mitigation.

Perturbing the original data and extracting prior knowledge from the original dataset are two commonly used methods for VSG [58]. In terms of perturbing the original data, Sun et al. [13] add Gaussian noise to the original samples to obtain new samples of

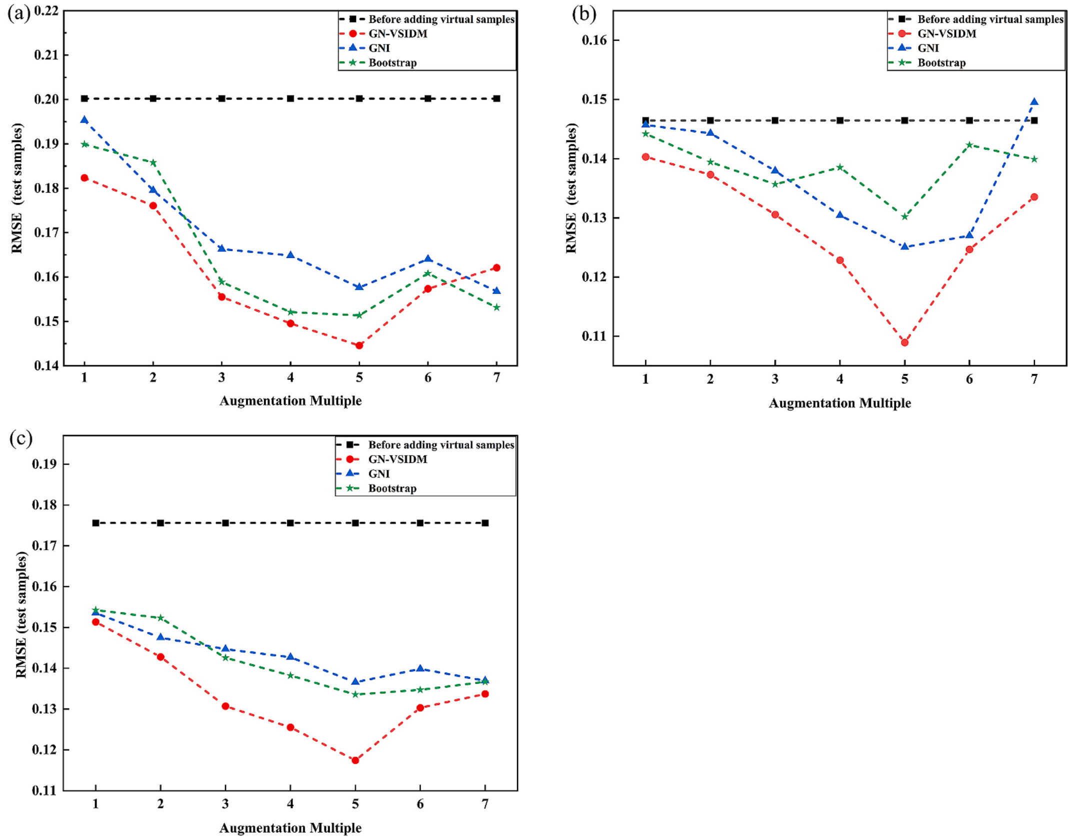

Fig. 12. The effect of different VSG methods on the performance of different models: (a) LSBoost, (b) XGBoost, and (c) RF.

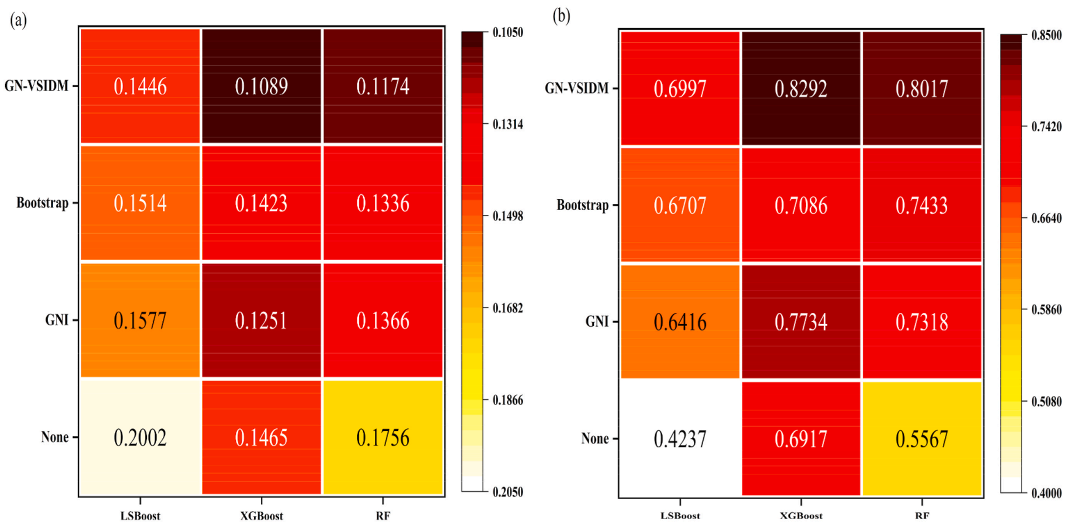

Fig. 13. The heatmap of joint models' performances: (a) RMSE and (b) R^2.

storm surge disaster losses. However, the parameters of Gaussian noise are directly given in Ref. [13] and there is some subjectivity in this method. While the GN-VSIDM introduces a hyperparameter SNR, which can generate Gaussian noise with suitable values in combination with original indicator data. Meanwhile, the optimal SNR is determined through comparative experiments, which increases the objectivity of the hyperparameter taking value. In terms of extracting prior knowledge, Zhao et al. [34] use the interpolation method to expand the direct economic loss samples of marine disasters. However, the interpolation method can only extract the definite information in the data, while the GN-VSIDM extracts the fuzzy information of the original dataset by the information diffusion technique. Therefore, the GN-VSIDM extracts more prior knowledge from statistics.

Table 4
Main parameters of the three MLMs.

LSBoost		XGBoost		RF	
Parameter	**Value**	**Parameter**	**Value**	**Parameter**	**Value**
min_leaf	8	learning_rate	0.3	n_estimators	100
learner_num	30	min_split_loss	0	min_samples_split	2
learning_rate	0.1	reg_lambda	1	min_samples_leaf	1
		reg_alpha	0	max_features	7
		max_depth	6		

Table 5
The comparison of the GN-VSIDM-XGBoost model with the XGBoost, BPNN, and DT models.

Model	GN-VSIDM-XGBoost	XGBoost	BPNN	DT
RMSE	0.1085	0.1465	0.1560	0.1604
R^2	0.8292	0.6917	0.6500	0.6286

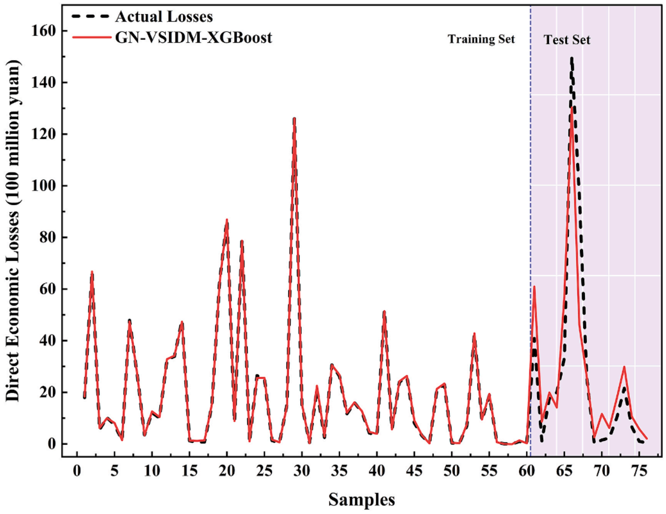

Fig. 14. The results of TSSDL estimation in the study area.

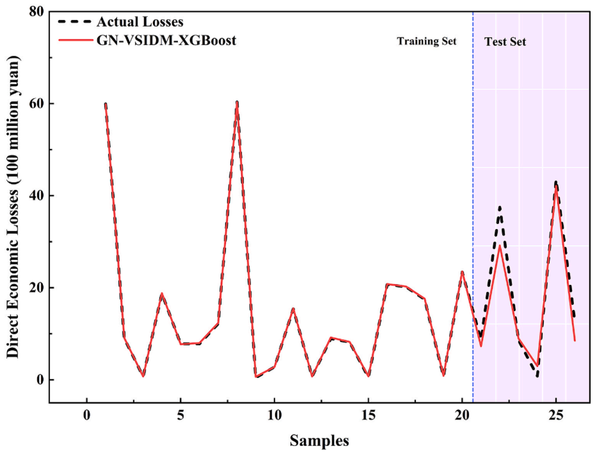

Fig. 15. The results of TSSDL estimation in Zhejiang Province.

In addition, the analysis of the method principle and experimental results in this paper is as follows.

(1) Superiority of the RF-RFECV method in indicator screening

The RF method has randomness in calculating the importance of indicators, which causes uncertainty in determining the optimal SoI. For this phenomenon, the RF-RFECV method uses the cross-validation method and the statistical results of multiple experiments to calculate the indicators' importance, which makes full use of the sample information and reduces the randomness of the RF method. As shown in Fig. 9, the SoI obtained by the RF-RFECV method has the advantage of few indicators and high importance. The RF-RFECV method provides a theoretical basis and practical application for eliminating the randomness of indicator importance calculation. More methods to eliminate the randomness of indicator importance calculation deserve more in-depth study.

(2) Parameters determination during virtual sample generation

Determination of the optimal Gaussian noise

Adding Gaussian noise to the indicators data for data perturbation during VSG is typical for sample augmentation [13,60]. However, adding improper Gaussian noise will reduce the quality of the generated virtual samples, making the model cannot achieve optimal performance. Currently, the mainstream method sets the Gaussian noise directly as a constant, which lacks pertinence and is unsuitable for various application scenarios.

The hyperparameter SNR is applied to calculate the optimal Gaussian noise value according to the characteristics of the original samples. Besides, the SNR value that minimizes the model error is determined quantitatively through multiple experiments, and the VSG results are satisfactory. However, this method is driven by extensive experiments and consumes a lot of time. Therefore, future research should explore theoretical formulas for calculating the optimal range of SNR values. Besides, the theoretical formulas should be established by profoundly studying the existing VSG principles.

Determination of the optimal augmentation multiples

Adding virtual samples to the original dataset shows excellent performance for solving the small sample problem with easy overfitting and weak generalization ability. In the current research on the VSG methods, scholars determine the optimal number of the added virtual samples by gradually increasing the number of virtual samples from small to large and comparing the corresponding model's performance (estimation error or classification accuracy). In general, as the number of virtual samples increases, the estimation model performance trends to be better and then worse [62,70]. Therefore, there is a threshold or range of the number of added virtual samples for optimal model performance.

This paper adopted the same idea to determine the optimal augmentation multiples and the estimation model achieved the optimal performance when the augmentation multiples are five. However, there is no unified and mature theoretical method to derive the optimal number of virtual samples. Therefore, it is valuable to establish an academic theory and formula to calculate the number of virtual samples reasonably.

(3) Estimation of outliers

According to the result of the TSSDL estimation, which is shown in Fig. 14, the estimated loss curve fits well with the actual loss curve. Especially, the actual loss of the 66th sample is significantly larger than the other samples' losses, and it is scarce among the training samples. This phenomenon is attributed to the outlier problem, which is common in disaster loss estimation, and the loss of the 66th sample is regarded as an outlier. The outlier problem is inevitable in disaster loss data, and removing the outlier data directly is a common method in current research [71,72]. In contrast, the outlier is retained in this paper. The GN-VSIDM-XGBoost model achieves comparatively good results for estimating the outlier, which is close to the maximum value of the training set. Moreover, this result indicates that the GN-VSIDM-XGBoost model shows excellent performance, and the GN-VSIDM method effectively solves the small sample problem of TSSDL estimation.

5. Conclusion

Due to the limitations of technical level, economic cost, and other factors, the number of storm surge disaster loss samples is small and the information is incomplete, which leads to the low accuracy and weak robustness of the TSSDL estimation models. To obtain a TSSDL estimation model with high accuracy and strong robustness under small sample conditions, this paper proposes a TSSDL estimation method based on the VSG method and MLMs. This method mainly contains the steps of comprehensive indicator system construction, indicator screening, sample augmentation based on the VSG method, optimal estimation model construction, and optimal model performance evaluation.

Firstly, based on the principle of comprehensiveness, a TSSDL indicator system containing 29 indicators in three aspects, i.e., the hazard of disaster-causing factors, the vulnerability of disaster-bearing bodies, and resilience, was constructed and it was verified that the indicator system was effective for estimating TSSDL. Secondly, based on the RF-RFECV indicator screening method, the optimal indicator subset containing 7 indicators was obtained and its cumulative importance reached 0.6739 of whole. The optimal indicator subset was verified to have the advantages of a small number of indicators, high cumulative importance, and strong interpretability. Besides, the RF-RFECV was justified in indicator screening. Thirdly, the experimental comparison results showed that the GN-VSIDM VSG method proposed in this paper was superior to the GNI and Bootstrap methods. In addition, the GN-VSIDM maked the TSSDL estimation models achieve the best performance under the condition that the SNR was 50 dB and the augmentation factor was 5 times. Therefore, the GN-VSIDM proposed in this paper effectively addressed the performance limitation of the MLMs due to the small sample problem. Then, the cross-comparison experimental results of four types of samples and three MLMs showed that the joint model

GN-VSIDM-XGBoost performed the best with the smallest RMSE of 0.1089 and the largest R^2 of 0.8292. Finally, based on the storm surge disaster samples data in the study area and the verification area, the optimal TSSDL estimation model was verified to have high accuracy and strong robustness.

This study provides a scientific basis for storm surge disaster mitigation decision-making and risk management. Moreover, this study can be used as a typical demonstration case to provide theories and methods for other studies with a small sample problem.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in the paper are collected from official open-source data, and the data acquisition method is given in the paper.

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